

Question description

Question No:9

Rathik organized technical round interview in Macrosoft for the set of computer science candidates.

The problem is to perform Implement a stack using single queue. you have to use queue data structure, the task is to implement stack using only given queue data structure.

Rathik have given the deadline of only 5 minutes to complete the problem.

Can you Help the candidates to complete the problem within the specified time limit ?

Function Description

x is the element to be pushed and s is stack

push(s, x)

- 1) Let size of q be s.
- 2) Enqueue x to q
- 3) One by one Dequeue s items from queue and enqueue them.

Removes an item from stack

pop(s)

1) Dequeue an item from q

Constraints

$0 < n, m < N$

$1 < arr[i] < 1000$

Input Format:

First line indicates n & m, where n is the number of elements to be pushed into stack and m is the number of pop operation need to be performed

next line indicates the n number stack elements

Output Format:

First line indicates top of the element of the stack

second line indicates the top of the element after the pop operation

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

```
5 2
1 2 3 4 5
```

EXPECTED OUTPUT

```
top of element 5
top of element 3
```

Test Case 2

INPUT (STDIN)

```
15 4
9 8 7 6 5 5 6 7 8 3 11 2 3 4 5
```

EXPECTED OUTPUT

```
top of element 5
top of element 11
```

Code:

```
1  #include <stdio.h>
2
3  #define MAX 100
4
5  int queue[MAX];
6  int front = 0;
7  int rear = 0;
8  int size = 0;
9
10 /* Enqueue */
11 void enqueue(int x)
12 {
13     queue[rear] = x;
14     rear = (rear + 1) % MAX;
15     size++;
16 }
17
18 /* Dequeue */
19 int dequeue()
20 {
21     int x;
22
23     if (size == 0)
24         return -1;
25
26     x = queue[front];
27     front = (front + 1) % MAX;
28     size--;
29 }
```

```
30     return x;
31 }
32
33 /* Push into stack using one queue */
34 void push(int x)
35 {
36     int oldSize;
37     int i;
38     int temp;
39
40     oldSize = size;
41
42     enqueue(x);
43
44     /* Move all previous elements behind x */
45     for (i = 0; i < oldSize; i++)
46     {
47         temp = dequeue();
48         enqueue(temp);
49     }
50 }
51
52 /* Pop from stack */
53 int pop()
54 {
55     return dequeue();
```

```
56     }
57
58     /* Peek */
59     int peek()
60     {
61         if (size == 0)
62             return -1;
63
64         return queue[front];
65     }
66
67     int main()
68     {
69         int n, m;
70         int x;
71         int i;
72
73         scanf("%d %d", &n, &m);
74
75         /* Push elements */
76         for (i = 0; i < n; i++)
77         {
78             scanf("%d", &x);
79             push(x);
80         }
81
82         /* Top before pop */
```

```

83     printf("top of element %d\n", peek());
84
85     /* Perform pop operations */
86     for (i = 0; i < m; i++)
87     {
88         pop();
89     }
90
91     /* Top after pop */
92     printf("top of element %d\n", peek());
93
94     return 0;
95 }

```

Output:

Test Case-1:

```

5 2
1 2 3 4 5
top of element 5
top of element 3

```

Test Case-2:

```

15 4
9 8 7 6 5 5 6 7 8 3 11 2 3 4 5
top of element 5
top of element 11

```

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